Context-Preserving Motion Differentials for Adversarial Motion Imitation

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Abstract— **Physics-based motion imitation requires a controller to match not only a reference pose, but also the motion that produced it. Paired tracking rewards usually subtract the simulated state from the reference before scoring the error. This preserves correspondence but removes shared motion context: two reference–policy pairs with the same differential become indistinguishable, even when they occur in different motion regimes. We introduce *context-preserving motion differentials* (CPMD), a causal representation that maintains a fixed-state causal motion summary for each side, applies the same horizon-independent quadratic encoding, and subtracts only afterward. The resulting differential preserves the exact zero-error target while exposing selected interactions between tracking error and common motion context. Training remains a single discriminator, a standard zero-versus-policy binary objective, and one scalar imitation reward. On a contact-rich humanoid Roll, a first-order history representation follows the forward displacement but produces almost no directed rotation and takes a shortcut in all 256 evaluations. Under the same training seed and sample budget, CPMD reaches 0.930 mean directed winding and reduces the shortcut rate to 5.9%. In this controlled run, the encoded representation recovers directed Roll while the first-order representation does not, exposing a behavior failure hidden by conventional tracking errors.**

I. Introduction

Learning physically simulated characters from motion capture requires two properties that are easy to conflate. The reward must preserve *correspondence*: the simulated character should reproduce the intended reference instant rather than any plausible pose. It must also preserve *motion context*: the meaning of an error depends on what the reference and character are doing around that instant. A small rotational discrepancy may be harmless while standing but decisive during a flip, roll, or recovery.

Reference-conditioned methods commonly obtain correspondence by comparing an aligned reference state with the simulated state. Hand-designed objectives map pose, velocity, and end-effector errors through separately tuned kernels; learned differential objectives instead give the complete tracking differential to a discriminator. Both strategies normally form the error first and evaluate it afterward. This order introduces a simple information bottleneck. If $x^r - x^\pi$ is the scorer input, then jointly shifting both operands by the same value leaves the input unchanged. A more expressive scorer can reshape the reward over observed differentials, but it cannot recover pair information that subtraction has removed.

The limitation is visible in periodic, contact-rich motions. Consider a forward Roll. A controller can reduce root-position and body-pose errors by lying down, sliding forward, and rising again without accumulating the reference's directed body rotation. The shortcut may survive for a complete episode and score well on mean tracking error. Generic fall termination is also unavailable: torso and hip contact are part of the desired behavior. The problem is therefore neither a simulator crash nor an invalid reference; it is a low-error trajectory with the wrong defining motion.

We change the order of two operations: encode each motion history first, then take the reference–policy difference. The evaluated instantiation summarizes root and joint increments causally in one fixed motion frame and applies the same cross-coordinate quadratic map to both summaries. This *encode-before-difference* order retains the familiar zero-centered target while revealing how a tracking differential interacts with motion common to the two sides. No reward branch, auxiliary temporal loss, manually weighted trajectory term, recurrent discriminator, or termination rule is introduced.

Our contributions are twofold:

- We identify premature differencing as an information bottleneck and introduce CPMD, an encode-before-difference representation that retains phase alignment and the exact zero-error target while exposing selected error–context interactions.
- Using directed winding and shortcut rate, we show a same-budget behavioral gap between first-order history and CPMD that is hidden by position error and episode length.

II. Related Work

Reference-conditioned tracking. DeepMimic established reinforcement-learning-based tracking with aligned pose, velocity, end-effector, and center-of-mass objectives [1]. UniCon and PHC scale such controllers to large motion collections and improve generalization or failure recovery [2], [3]. KungfuBot/PBHC adapts tracking tolerances through bilevel optimization [4], while HIL combines tracking and adversarial imitation for athletic control [5]. SimPoE couples simulated control to video-conditioned pose estimation [6], and H2O transfers motion imitation to real humanoid teleoperation [7]. These systems advance controller scale, robustness, conditioning, or deployment. Our concern is complementary: the information retained by a synchronized reference–policy representation before it is scored. Learned differential discriminators already replace a manually aggregated tracking reward with a nonlinear score over a zero-centered paired error [8]. We retain the adversarial objective and scalar reward mapping while altering the representation they receive.

Motion priors and reusable representations. GAIL learns rewards through occupancy matching [9]. MCP composes learned primitives, and NPMP compresses expert policies

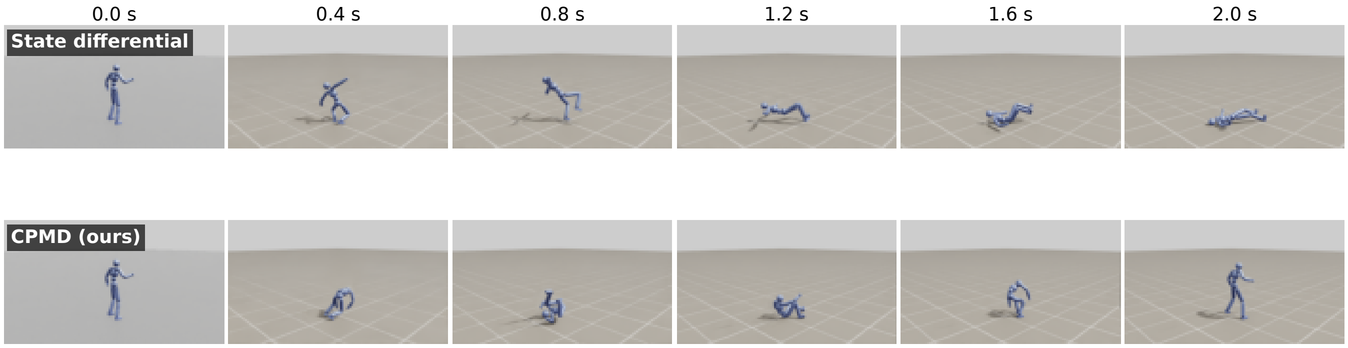

Fig. 1. A low-error trajectory need not reproduce the defining behavior. The instantaneous differential baseline follows part of the forward motion but accumulates almost no directed Roll. CPMD exposes error–context interactions absent from a difference-first input; this rollout rejects the shortcut. Both sequences use a phase-zero reset, deterministic policy, identical camera, and a 2 s horizon.

into a reusable latent motor module [10], [11]. AMP learns adversarial style priors from unstructured clips [12]; ASE and CALM extend this paradigm with reusable or directable embeddings [13], [14]. PULSE distills a large motion imitator into a universal latent space [15], MaskedMimic formulates versatile control as masked motion inpainting [16], and ADP uses temporal dynamics features without explicit phase-aligned pose tracking [17]. These methods capture broad behavioral context. We instead retain synchronized correspondence to one prescribed motion and change only how the paired histories are differenced.

Motion representations. Continuous rotation coordinates and exponential maps provide stable local geometry for articulated motion learning [18], [19]. Bilinear maps expose pairwise feature interactions in a variety of representation-learning settings [20]. We use a fixed stateless quadratic map for a specific purpose: encode each causal motion summary before taking the reference–policy difference.

III. Method

A. Differential Adversarial Tracking

Conventional tracking rewards manually aggregate terms such as pose, velocity, and end-effector error. Differential adversarial tracking instead places the complete synchronized tracking differential at the input of a learned discriminator [8]:

$$\begin{aligned}\Delta_t^s &= \phi(\hat{s}_t) \ominus \phi(s_t),\\ \mathcal{J}_D &= \tfrac{1}{2}\log D_\theta(0) + \tfrac{1}{2}\mathbb{E}_{\Delta\sim q_\pi}\log[1-D_\theta(\Delta)] - \lambda_{\mathrm{GP}}\mathcal{L}_{\mathrm{GP}},\\ r_t &= -\beta\log[1-D_\theta(\Delta_t^s)]\,.\end{aligned} \tag{1}$$

Here ϕ constructs the synchronized state features, $\ominus$ denotes the appropriate coordinate-wise difference, and the discriminator maximizes $\mathcal{J}_D$. The distribution q_π is the empirical mixture of current and replay policy differentials used by our shared implementation. The exact match remains the single positive sample at the origin. Equation (1) is the learning backbone retained in all of our comparisons; our method changes only the differential supplied to it.

This elegant zero-centered construction also makes its information boundary precise. Let x^r and x^π be any additive coordinate block of the aligned features, with $\Delta_x(x^r, x^\pi) = x^r - x^\pi$. For every common shift c,

$$\Delta_x(x^r + c, x^\pi + c) = x^r - x^\pi = \Delta_x(x^r, x^\pi). \tag{2}$$

Consequently, every scorer $f(\Delta_x)$ is constant over the entire family $(x^r + c, x^\pi + c)$. Network depth and width cannot distinguish members of this family because they are represented by exactly the same input. In motion imitation, c corresponds to a common operating regime: the same tracking error can occur while both trajectories are nearly stationary or while both carry large coordinated motion.

We therefore alter the representation while retaining the adversarial training objective and scalar reward mapping: apply the same causal operator to each side, encode both outputs, and subtract only afterward. The construction must remain exactly zero under perfect tracking and use no future states.

B. A Causal Motion Summary

For side $a \in \{r, \pi\}$, let p_t^a, R_t^a, and Q_t^a denote root position, root orientation, and articulated joint rotations. The evaluated summary uses the same developed increment and discounted accumulator on both sides:

$$\begin{aligned}\xi_t^a &= \begin{bmatrix} C_0(p_t^a - p_{t-1}^a) \\ C_0\,\mathrm{Log}(R_t^a(R_{t-1}^a)^{-1}) \\ \mathrm{Log}_{\mathcal{J}}((Q_{t-1}^a)^{-1}Q_t^a)\end{bmatrix} \in \mathbb{R}^{d_\xi},\\ m_t^a &= e^{-\Delta t/\tau}m_{t-1}^a + \xi_t^a, \qquad m_0^a = 0.\end{aligned} \tag{3}$$

C_0 is the inverse heading of the reference at phase zero and remains fixed for the episode. Root translation and spatial rotation therefore share one motion-anchored frame even when the character later becomes horizontal or inverted. $\mathrm{Log}_{\mathcal{J}}$ maps each joint increment to tangent coordinates, using the joint axis for hinges and exponential coordinates for spherical joints. Quaternion representatives are selected on the shortest branch before taking the logarithm. These are per-control-step displacements, not finite differences of an already mixed observation vector.

Using one fixed anchor is important for Roll. Recomputing heading from the current root orientation becomes ill-conditioned near inversion and can inject artificial sign changes. The fixed anchor is also equivariant to a common world translation and heading rotation: recomputing it from

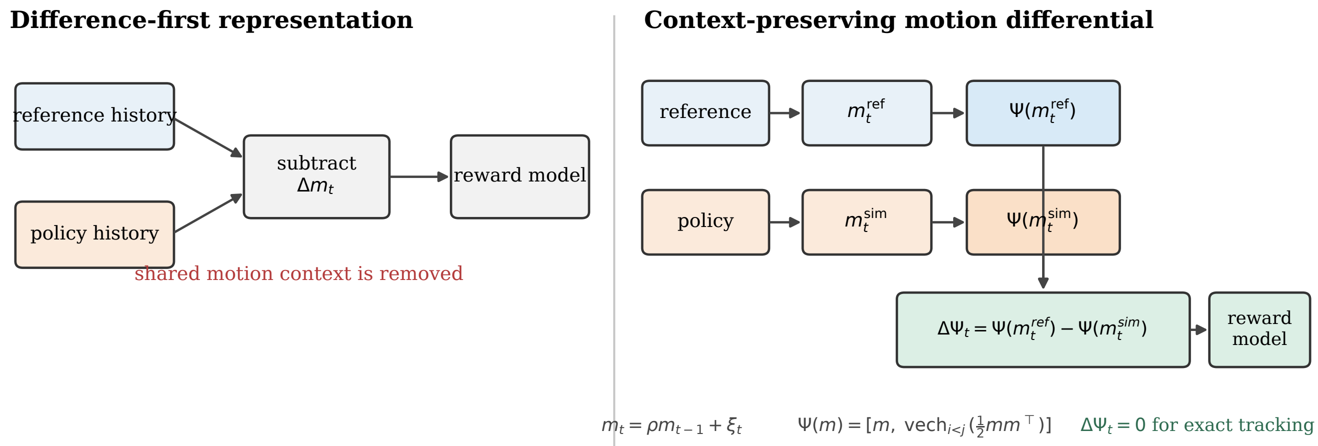

Fig. 2. Difference-first objectives remove common-mode information before reward learning. CPMD summarizes and encodes each history in the same fixed frame, then subtracts. The training objective and scalar reward remain unchanged.

the transformed reference gives identical developed increments.

Here τ is the sole memory-scale parameter, expressed in seconds. The decay assigned to a physical lag ℓ is $e^{-\ell/\tau}$ regardless of control frequency. The implementation stores only m_t^r and m_t^π; it does not retain a sequence buffer or recurrent second-order state.

At reset, both accumulators are zeroed and both previous poses are initialized from the same aligned reference state. The first nonzero increment is therefore the first genuine post-reset transition, and partial resets modify only the corresponding parallel environments. For cyclic motion, unwrapped root displacement is used across the reference boundary so that wrapping does not create a false teleport.

C. Context-Preserving Differential

We encode each summary with its linear coordinates and one copy of every cross-coordinate quadratic interaction, then append the history differential to the instantaneous differential in Eq. (1):

$$\begin{aligned} \Psi(m) &= \left[m; \Pi_<\left(\tfrac{1}{2}mm^\top\right)\right], \\ \widetilde{\Delta}_t &= \left[\Delta_t^s; \Psi(m_t^r) - \Psi(m_t^\pi)\right]. \end{aligned} \tag{4}$$

Here $\Pi_<(M) = \{M_{ij}\}_{i<j}$ denotes the strict upper triangle. The linear component ensures that the history block is zero if and only if $m_t^r = m_t^\pi$. The quadratic component does not create a competing target; perfect tracking still produces $\widetilde{\Delta}_t = 0$ exactly.

The extra information is visible in closed form. Define $\Delta m = m^r - m^\pi$ and $\Sigma m = m^r + m^\pi$. The quadratic differential is

$$\Delta S = \tfrac{1}{4}\Pi_<\left(\Delta m\, \Sigma m^\top + \Sigma m\, \Delta m^\top\right). \tag{5}$$

Equation (5) follows by expanding the two outer products. It shows that CPMD exposes pairwise interactions between the tracking differential and the motion common to reference and policy. Two pairs may share the same Δm while producing different ΔS because their Σm differs. The scorer can therefore condition an error on the motion regime in which it occurred. Here "context-preserving" has a specific scope: selected common-mode interactions survive differencing. The representation is not a lossless encoding of either trajectory or of the complete common summary.

a) A two-dimensional example.: The information difference does not require a complicated trajectory. Let a reference summary be $(1, 0)$ and a simulated summary be $(0, 0)$. Their first-order differential is $(1, 0)$, while their only strict-upper quadratic coordinate has zero differential. Now add the same context $(0, 1)$ to both summaries. The new pair is $(1, 1)$ and $(0, 1)$: its first-order differential remains exactly $(1, 0)$, but its quadratic differential becomes $1/2$. A difference-first scorer must assign both pairs the same input. CPMD can distinguish them because the same error now co-occurs with motion in the second channel. The humanoid construction uses this mechanism across root and joint increments rather than abstract coordinates.

The evaluated map packs each unordered cross-coordinate product once and omits self-products. This indexing is an implementation choice, not a separate methodological claim; the encode-before-difference principle does not require dropping the diagonal. For the humanoid used here, $d_\xi = 3+3+28 = 34$, giving $d_\xi(d_\xi - 1)/2 = 561$ cross-coordinate terms. Together with the 172-D instantaneous differential, the discriminator input is 767-D. The recurrent state remains two 34-D vectors; the quadratic block is materialized statelessly when the current observation is constructed.

Table I separates temporal memory from contextual encoding. L1 already uses the same causal accumulator, so the L1–CPMD comparison does not merely ask whether history helps. It asks whether independently encoding the two histories before differencing changes the learned objective. The comparison does not hold input width fixed; Sec. IV-F states the matched-width control needed for that stronger conclusion.

TABLE I

INFORMATION EXPOSED BY THE EVALUATED REPRESENTATIONS. "CONTEXT" MEANS SELECTED DEPENDENCE ON THE COMMON SUMMARY Σm, NOT LOSSLESS RECOVERY OF THAT SUMMARY.

Input	State	History	Context	Dim.
Instantaneous	✓	–	–	172
First-order history	✓	✓	–	206
CPMD	✓	✓	✓	767

D. One Differential, One Reward

We substitute $\widetilde{\Delta}_t$ from Eq. (4) for Δ_t^s in Eq. (1); nothing else is added to the policy objective. With $D_\theta = \sigma(z_\theta)$, the implemented reward is the nonnegative capped form $\beta \min\{\mathrm{softplus}(z_\theta(\mathcal{N}(\widetilde{\Delta}_t))), -\log 10^{-4}\}$. Away from the numerical cap, this is exactly the reward in Eq. (1) after the stated replacement and normalization. Thus training still has one exact-zero positive, one negative batch formed from current and replay differentials, one feed-forward discriminator, and one scalar imitation reward. There is no separate trajectory reward, auxiliary branch loss, or modified termination.

The implementation evaluates D_θ on a scale-normalized differential $\mathcal{N}(\widetilde{\Delta}_t)$, where $\mathcal{N}(0) = 0$ and no running mean is subtracted. The original formulation regularizes discriminator gradients at policy differentials [8]; our shared implementation instead uses a two-endpoint *logit*-gradient penalty at the zero anchor and sampled policy differentials, plus an ℓ_2 penalty on the final logit weights. This fixed variant is identical in L1 and CPMD and is not part of the proposed representation.

The classifier objective should not be confused with a behavior metric. Perfect positive/negative classification only states that zero and the current policy distribution are separable; it does not show that the reward varies smoothly over policy-reachable states or that the target motion has been completed. We therefore evaluate behavior with winding and inspect policy reward statistics in addition to discriminator accuracy.

a) Correctness checks.: The implementation follows the control-step boundary explicitly. Immediately before physics it stores the previous reference and simulated poses. After physics, the reference advances to the same new time as the simulator and one increment is pushed to each accumulator. The history differential is copied into the rollout record before reset modifies environment buffers. This ordering avoids both a one-step reference offset and a false transition between episodes.

Correctness tests verify that perfect tracking gives exact zero, the history encoding is invariant to common world translation and heading, q and $-q$ give identical increments, partial resets do not alter other environments, and Eq. (5) matches the indexed implementation. These checks establish formula–code agreement; policy quality is evaluated independently.

IV. EXPERIMENTS

A. Setup

Simulation and training. Experiments use a 15-body humanoid in Isaac Lab [21] at 30 Hz control frequency. The actor and critic are two-layer MLPs with 1024 hidden units; the discriminator is a 1024–512 MLP. Policies are optimized with PPO [22] using 4096 parallel environments, 32-step rollouts, discount 0.99, GAE 0.95, and a 131.072M-sample budget. The discriminator uses gradient-penalty weight 2, logit regularization 0.01, and a replay buffer. Differential normalization tracks per-coordinate absolute scale without mean subtraction and freezes after 10^8 samples.

The reference-conditioned actor sees target frames at the next three control steps in every compared arm. Random phase reset is enabled. All representation comparisons use the same controller, optimizer, reward mapping, sample budget, and seed; only the discriminator input differs.

Roll task. The reference lasts 2.0 s and contains one full forward Roll with 0.902 m root displacement. Training episodes contain one cycle (60 control steps), while deterministic evaluation lasts 10 s (five cycles, 300 steps). Pose termination is disabled and all body contacts are allowed. This is necessary for the intended ground maneuver, but it also means episode length is always the configured horizon and cannot serve as success.

Compared representations. The primary controlled comparison is between: (1) a first-order causal input $[\Delta_t^s; m_t^r - m_t^\pi]$ (L1), and (2) CPMD from Eq. (4). Both use $\tau = 32/30$ s and the unchanged objective in Eq. (1). We additionally show an instantaneous state-differential checkpoint as a qualitative reference. That checkpoint was trained with a shorter 65.5M-sample budget, so it is not used for an equal-budget causal claim.

Protocol integrity. L1 and CPMD were launched from fresh initialization with the same random seed, 4096 environments, one-cycle training horizon, engine configuration, and normalizer schedule. The policy observation is unchanged, so the new history is available only to the discriminator and reward. Both use the nonnegative reward in Eq. (1); neither receives a handcrafted winding term, absolute tracking penalty, or failure bonus. With contact and pose termination disabled, the policy cannot improve return merely by ending a bad Roll early. These controls make L1 versus CPMD a clean test of the history representation, subject to the input-width limitation discussed below.

B. Behavior-Level Evaluation

Framewise pose error cannot determine whether a periodic path was traversed. We therefore integrate shortest-branch root-rotation increments about a fixed Roll axis. If $\delta\theta_k^a$ is the projected increment for reference or policy side a, directed winding is

$$w = \frac{\sum_{k=1}^{T} \delta\theta_k^\pi}{\sum_{k=1}^{T} \delta\theta_k^r}. \tag{6}$$

Incremental integration is essential because the orientations before and after a full 2π rotation are nearly identical. We classify an evaluation as a shortcut when $w < 0.5$ and retain the continuous winding value. We also report forward-displacement ratio, root-height RMSE, and landing-window upright alignment. Each history checkpoint is evaluated for 256 deterministic 10 s episodes; the qualitative instantaneous checkpoint has 128 evaluations.

The evaluation axes are fixed from the reference rather than inferred from the policy. The forward axis is the normalized reference root displacement over one 2 s cycle, and the Roll axis is its horizontal perpendicular. Every increment is projected onto that same axis, including when the policy is inverted. This prevents the metric from changing its meaning when the character heading becomes unreliable. Root-height RMSE is measured at every post-step sample. Upright alignment is evaluated only in the final 0.2 s landing window, because averaging uprightness over a correct Roll would incorrectly penalize its inverted middle phase.

The threshold $w < 0.5$ is used as a transparent diagnostic division between the near-zero and near-reference modes visible in Fig. 4; it is not used by the policy. A final multi-seed study should freeze the threshold before training and report a threshold-sensitivity curve. Our main evidence therefore remains the continuous winding distribution, with shortcut rate serving as an interpretable count of its low-winding mode.

C. Does Context Recover the Defining Motion?

Table II exposes two qualitatively different solutions. L1 follows forward motion: its displacement ratio is 1.085 and its mean root-height error is only 6.1 cm. Conventional position and height metrics would therefore make this run look successful. Yet its mean winding is 0.035, its median is slightly negative, and every evaluation is a shortcut. The controller has learned to move with the reference without executing the directed rotation.

Under the same training budget and seed, CPMD reaches 0.930 mean winding and a median of 0.976; 241 of 256 evaluations cross the diagnostic winding threshold. Root-rotation and root-angular-velocity errors also decrease by 43% and 38% relative to L1. The slightly higher height RMSE is consistent with actually entering the low and inverted configurations of the reference, rather than remaining in an easier near-ground shortcut. Figure 3 shows that the difference is not an artifact of episode length: all methods reach the fixed horizon, while their winding distributions occupy distinct behavior modes.

The qualitative rollout in Fig. 1 matches the metrics. The instantaneous policy lowers its body and translates but does not complete the turn. CPMD enters the contact phase, passes through inversion, and returns upright. Since both use identical target observations and legal contacts, the distinction is attributable to what their reward representation can expose, not to privileged future input or a hand-coded fall rule.

D. Distribution and Failure Anatomy

Figure 4 shows why the mean alone is incomplete. For L1, the 10th, 50th, and 90th winding percentiles are -0.017, -0.005, and 0.169: even its upper tail does not approach one full reference Roll. For CPMD, the corresponding values are 0.954, 0.976, and 1.017. The lower mean of 0.930 is caused by a distinct set of 15 low-winding evaluations rather than a broad tendency to under-rotate. We therefore report both continuous winding and the number of low-winding outcomes.

The landing-upright indicator is 30.5% for both L1 and CPMD. Thus the current evidence supports recovery of directed rotation, not a claim that the controller has solved every landing and stabilization failure. This distinction matters: winding tests whether the defining path occurred, while upright alignment tests a different terminal property. A controller may pass one and fail the other. The final benchmark should report them separately rather than merging both into one opaque success label.

E. Training Dynamics

Figure 5 reports diagnostics at matched environment samples. Both history methods initially reduce easy translational and posture errors. L1 continues improving body-position error and eventually reaches a low root position error, but its root-rotation and angular-velocity curves plateau at a non-rolling solution. CPMD undergoes a later transition in the rotational quantities and finishes with substantially lower errors. This behavior explains why a short early-training comparison is misleading: the defining Roll emerges after the policy has learned enough contact control to cross the rotational barrier.

The discriminator itself remains a standard zero-versus-policy classifier. No trajectory-success labels, winding reward, or manually selected Roll axis enter training; winding is used only for evaluation. Thus the behavior is not obtained by directly rewarding the reported metric. The quadratic block instead makes motion-regime interactions available to the learned reward.

F. Representation and Computational Cost

L1 and CPMD share the same 34-D recurrent state per side. L1 yields a 206-D discriminator input; CPMD adds 561 stateless cross-channel coordinates for a 767-D input. The discriminator grows from 0.737M to 1.312M trainable parameters, while the actor and critic remain identical. Measured throughput changes from 55,159 to 54,165 samples/s, a 1.8% reduction. This is considerably cheaper than maintaining a sequence window or recurrent second-order state, although rollout and replay storage increase with the larger discriminator observation.

The current experiment isolates first-order history from the final context-preserving representation, but it does not by itself separate exposed common context from every effect of input width and polynomial conditioning. A decisive final control is a 767-D difference-first input $[\Delta^s; \Delta m; \Pi_<(\Delta m \Delta m^\top/2)]$. It has the same width and quadratic degree but remains a function of Δm alone and is therefore invariant to the common shifts in Eq. (2). This control, and paired training

TABLE II

ROLL EVALUATION AND FINAL TRAINING DIAGNOSTICS. L1 AND CPMD USE THE SAME SEED, 131.072M SAMPLES, AND 256 TEN-SECOND EVALUATIONS. THE INSTANTANEOUS CHECKPOINT IS A SHORTER-BUDGET QUALITATIVE REFERENCE. LOWER IS BETTER FOR ERRORS AND SHORTCUT RATE; HIGHER IS BETTER OTHERWISE.

Method	Train samples	Eval.	Wind. mean	Wind. med.	Shortcut	Disp. ratio	Height RMSE	Upright	Root rot.	Root ang. vel.	Samples/s
Instantaneous differential	65.5M	128	−0.007	−0.006	100.0%	0.221	0.343	0.0%	1.442	1.765	57,657
First-order history (L1)	131.1M	256	0.035	−0.005	100.0%	1.085	**0.061**	30.5%	0.475	1.522	**55,159**
CPMD	131.1M	256	**0.930**	**0.976**	**5.9%**	**1.143**	0.082	30.5%	**0.270**	**0.951**	54,165

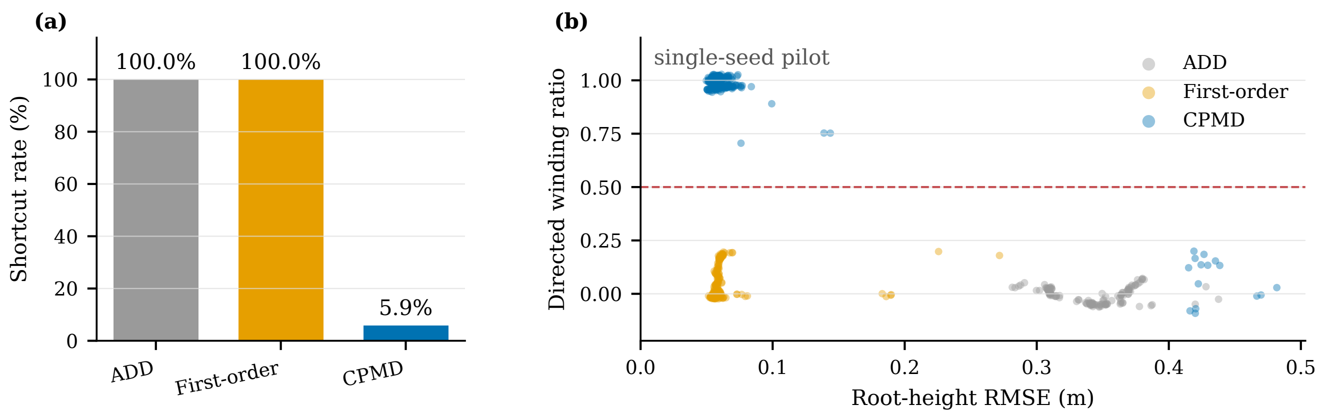

Fig. 3. Behavior-level evaluation reveals the non-rolling mode hidden by framewise errors. Bars show shortcut rate; points show directed winding against root-height RMSE. The instantaneous checkpoint is qualitative because its training budget differs. L1 and CPMD are the controlled, equal-budget comparison.

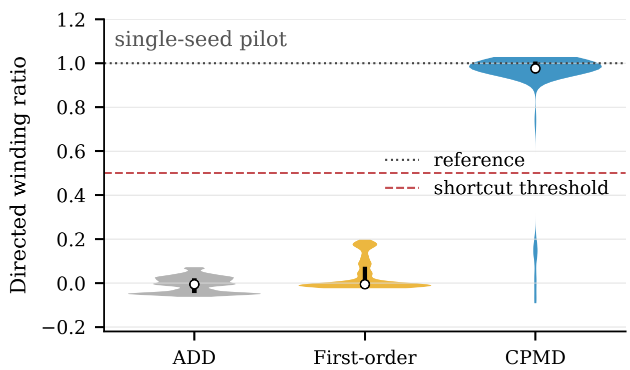

Fig. 4. Distribution of directed winding over the stored deterministic evaluations. L1 concentrates near zero. CPMD is concentrated near the reference value one, with a small separate failure mode. The instantaneous checkpoint is shown only as a qualitative reference.

seeds, are required before making a capacity-independent causal claim.

G. What the Current Evidence Establishes

Table III separates observations from conclusions that require additional experiments. This distinction is especially important because the 256 evaluation episodes share one trained policy. They measure variation over evaluation conditions, not the variance of the learning algorithm.

Two completed observations are nevertheless decisive for the stated pilot question. First, a low state error and accurate displacement do not imply the reference Roll occurred: L1 directly demonstrates that combination. Second, adding the independently encoded quadratic block changes the learned solution under one otherwise aligned training run. Neither observation requires calling the method state of the art or claiming that input capacity has been fully controlled.

TABLE III

EVIDENCE BOUNDARY OF THE COMPLETED ROLL STUDY.

Question	Current evidence
Does causal history alone induce Roll?	No in seed 0: L1 has 100% low-winding outcomes.
Does CPMD avoid that mode?	Yes in seed 0: 241/256 exceed $w = 0.5$.
Does it solve stable landing?	Not shown: upright rate remains 30.5%.
Is the gain independent of width?	Pending matched-width subtract-first control.
Is it robust across training?	Pending paired multi-seed runs.
Is it general across motions?	Pending easy and contact-rich tasks.

The mechanism suggested by Eq. (5) is also narrower than generic "long-term reasoning." CPMD does not store the full sequence, infer a symbolic action phase, or supervise path order. Its accumulator summarizes recent developed motion, and the quadratic block lets the current mismatch be evaluated relative to that common summary. For Roll, this exposes whether root and joint discrepancies occur while both sides carry the coordinated motion of the maneuver or while the character follows a low-motion shortcut. The experiment supports the usefulness of this context; it does not claim a lossless trajectory representation.

H. Final Statistical Protocol

The paper-ready comparison will treat the training seed, rather than an environment or evaluation episode, as the statistical unit. Each primary arm will use paired seeds,

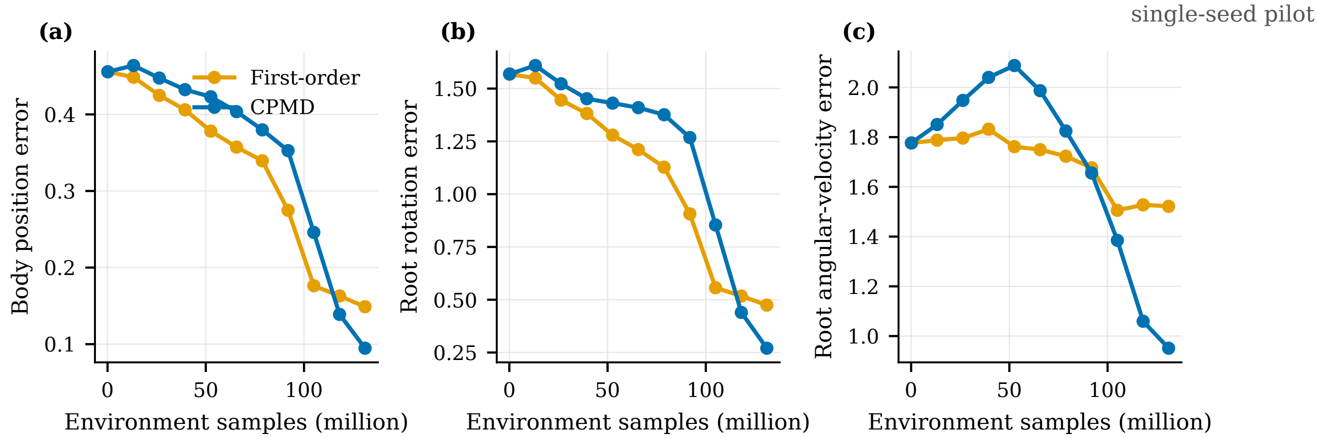

Fig. 5. Training diagnostics versus environment samples. First-order history improves several framewise quantities while remaining in the non-rolling mode. CPMD exhibits a late improvement in root rotation and angular velocity that coincides with the behavioral Roll solution. Curves show one paired training seed and are not confidence intervals.

identical initialization protocol, the same 131.072M sample budget, and the same checkpoint-selection rule. We will report seed-level mean and standard deviation, paired effect sizes, and confidence intervals for winding, shortcut rate, pose errors, and throughput. Evaluation conditions, threshold, phase/reset schedule, and perturbations will be fixed before launching the final runs.

The minimum primary matrix contains four representations: instantaneous state, L1, the matched-width subtract-first quadratic control, and CPMD. A conventional motion already solved by the instantaneous representation checks that added context does not degrade easy tracking; a second contact-rich, path-dependent motion checks whether the Roll result transfers. These runs complete the causal and generality claims without changing the method or introducing another reward component.

I. Scope and Limitations

The present results establish a large within-seed mechanism result on one motion, not universal superiority. Evaluation episodes from one checkpoint measure conditional behavior variation; they do not replace independent training seeds. A final statistical study must repeat the instantaneous, matched-width, L1, and CPMD arms over paired seeds at the same budget. It must also include an action already solved by the simpler representation to test non-degradation and another contact-rich path-dependent action to test breadth.

The representation has structural limits. The discounted accumulator can alias distinct old histories through cancellation, and the strict-upper map omits diagonal self-interactions. The $O(d_\xi^2)$ lifted width may become costly for much larger motion summaries. Finally, a better reward representation does not guarantee that finite-budget PPO will discover every dynamically difficult maneuver. These limitations concern the scope of the method, not the Roll metric: directed winding directly verifies whether the defining rotation occurred.

V. Conclusion

We presented context-preserving motion differentials, an objective-preserving encode-before-difference representation for adversarial motion imitation. By encoding reference and simulated histories independently, the method keeps phase-aligned correspondence while exposing selected interactions between tracking error and the shared motion regime. It changes neither the adversarial objective nor the scalar policy reward. In a controlled Roll experiment, first-order history produced accurate displacement but no directed rotation, whereas CPMD avoided the low-winding shortcut in 94.1% of evaluations. More broadly, the result shows why behavior-level path metrics and representation-level context are both necessary when low framewise error can hide the wrong motion.

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